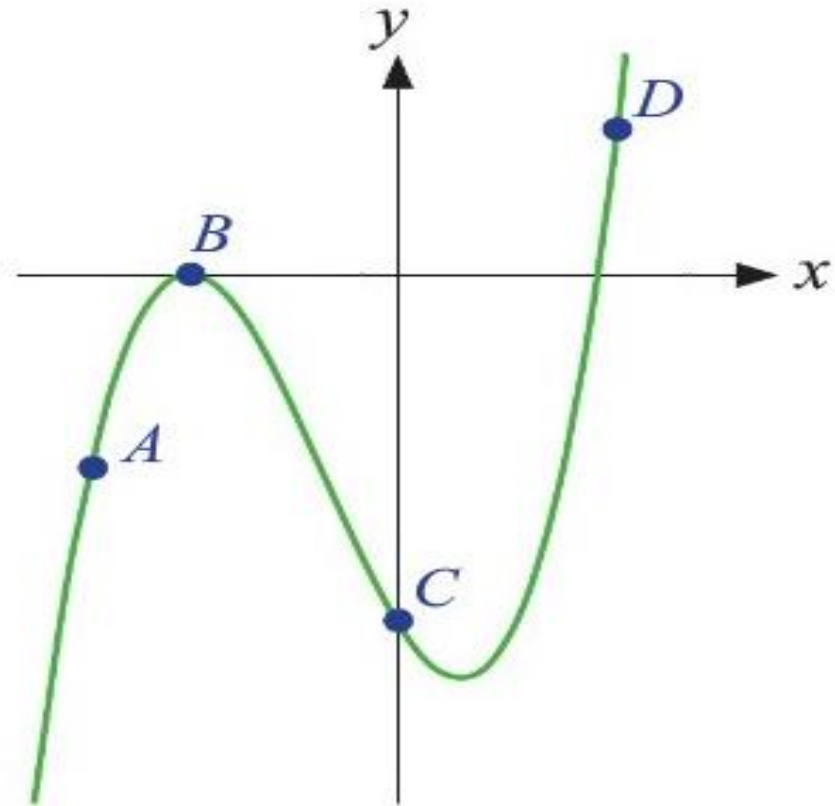


Chp22 微分及其应用(一)

Pg. 38

22.1 导数 Derivative

右图所示，绘出曲线上
点 A, B, C, D 处的切线。





切线的斜率

复习：二元一次直线方程式

$$\text{斜率 } m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

左图中，相异两点P,Q连线形成割线PQ

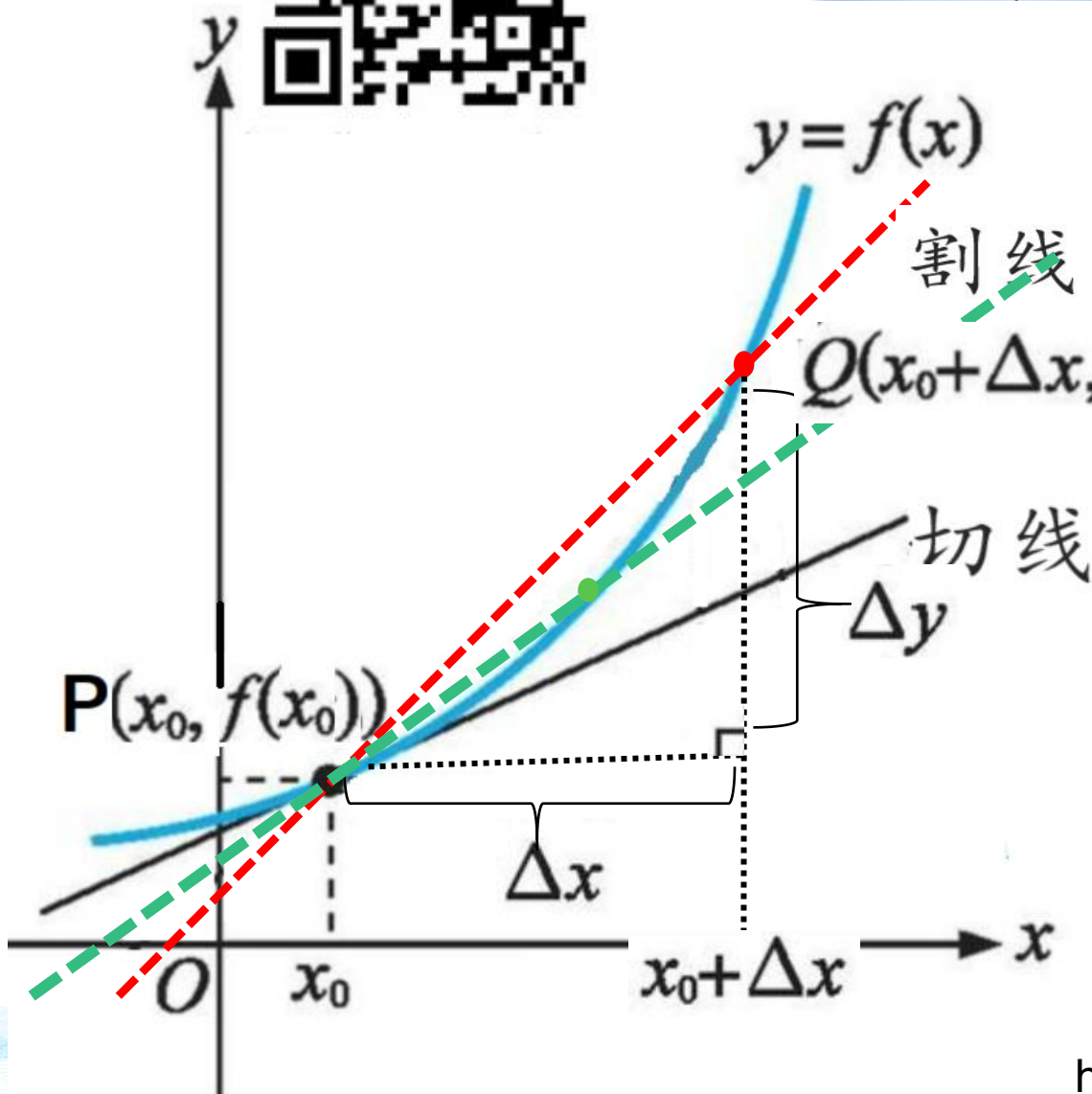
$$\begin{aligned}\text{则割线 } PQ \text{ 的斜率为 } \frac{\Delta y}{\Delta x} &= \frac{f(x_0 + \Delta x) - f(x_0)}{(x_0 + \Delta x) - x_0} \\ &= \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}\end{aligned}$$

当Q点沿着曲线无限接近P点(即 $\Delta x \rightarrow 0$)

则割线PQ的极限位置就是P点处的切线，

故P点处的切线斜率为

$$m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$



例)求曲线 $f(x) = x^2 + 1$ 在点 $P(1, 2)$ 处的切线的斜率。

设曲线 $y = f(x)$ 在点 $P(x_0, y_0)$ 处的切线斜率为

$$\begin{aligned} m &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[(x_0 + \Delta x)^2 + 1] - [(x_0)^2 + 1]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[x_0^2 + 2x_0\Delta x + \Delta x^2 + 1] - [(x_0)^2 + 1]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2x_0\Delta x + \Delta x^2}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} (2x_0 + \Delta x) \\ &= 2x_0 \end{aligned}$$

在点 $P(1, 2)$ 处的切线斜率 $m = 2(1) = 2$

即 $x_0 = 1$

曲线 $y = f(x)$ 在点 $P(x_0, f(x_0))$ 处的切线斜率为

$$m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

a) 求曲线 $f(x) = x^3$ 在点 $P(2, 8)$ 处的切线的斜率。

设曲线 $y = f(x)$ 在点 $P(x_0, y_0)$ 处的切线斜率为

$$\begin{aligned} m &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{(x_0 + \Delta x)^3 - (x_0)^3}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{x_0^3 + 3x_0^2\Delta x + 3x_0\Delta x^2 + \Delta x^3 - x_0^3}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{3x_0^2\Delta x + 3x_0\Delta x^2 + \Delta x^3}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} 3x_0^2 + 3x_0\Delta x + \Delta x^2 \\ &= 3x_0^2 \end{aligned}$$

在点 $P(2, 8)$ 处的切线斜率 $m = 3(2)^2 = 12$

即 $x_0 = 2$

曲线 $y = f(x)$ 在点 $P(x_0, f(x_0))$ 处的切线斜率为

$$m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

c) 求曲线 $f(x) = \frac{1}{\sqrt{x}}$ 在点 $P\left(4, \frac{1}{2}\right)$ 处的切线的斜率。

设曲线 $y = f(x)$ 在点 $P\left(4, \frac{1}{2}\right)$ 处的切线斜率为

$$\begin{aligned} m &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(4 + \Delta x) - f(4)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\frac{1}{\sqrt{4 + \Delta x}} - \frac{1}{2}}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2 - \sqrt{4 + \Delta x}}{2\Delta x \sqrt{4 + \Delta x}} \times \frac{2 + \sqrt{4 + \Delta x}}{2 + \sqrt{4 + \Delta x}} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\cancel{4} - (\cancel{4} + \Delta x)}{2\cancel{\Delta x} \sqrt{4 + \Delta x} (2 + \sqrt{4 + \Delta x})} \\ &= \lim_{\Delta x \rightarrow 0} \frac{-1}{2\sqrt{4 + \Delta x} (2 + \sqrt{4 + \Delta x})} = \frac{-1}{2\sqrt{4} (2 + \sqrt{4})} = -\frac{1}{16} \end{aligned}$$

曲线 $y = f(x)$ 在点 $P(x_0, f(x_0))$ 处的切线斜率为

$$m = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

设函数 $y = f(x)$ 在点 x_0 及其附近有定义, 当自变量 x 在点 x_0 处有增量 Δx 时, 则函数有相应的增量 $\Delta y = f(x_0 + \Delta x) - f(x_0)$

当 $\Delta x \rightarrow 0$ 时, Δy 与 Δx 之比的极限: $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ 存在,

则 $y = f(x)$ 在点 x_0 处可导或可微 (*Differentiable*), 称 $y = f(x)$ 在点 x_0 处的导数 (*derivative*),

记作 $f'(x_0)$ 或 $\left. \frac{dy}{dx} \right|_{x=x_0}$, 即 $f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$

若上述极限不存在。则称 $y = f(x)$ 在点 x_0 处的不可导或不可微。

$f(x)$ 的导函数 (简称导数) 为 $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

注：(1) 导函数简称导数，如不指明求某一点处的导数，则“求导数”即指“求导函数”

(2) 函数 $y = f(x)$ 的导函数 $f'(x)$ 与函数 $y = f(x)$ 的导数区别为 $f'(x)$ 是 x 的函数，

而 $f'(x_0)$ 是一个数值，其之间联系为 $f(x)$ 在点 x_0 处的导数 $f'(x_0)$ 就是导函数 $f'(x)$ 在点 x_0 处的函数值。

(3) 导数符号：
$$f'(x) = \frac{d}{dx} f(x) = \frac{df(x)}{dx},$$

又因 $y = f(x)$ ，所以 $f'(x) = y' = \frac{d}{dx} y = \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$ ，其中 $\frac{dy}{dx}$ 不是 $dy \div dx$

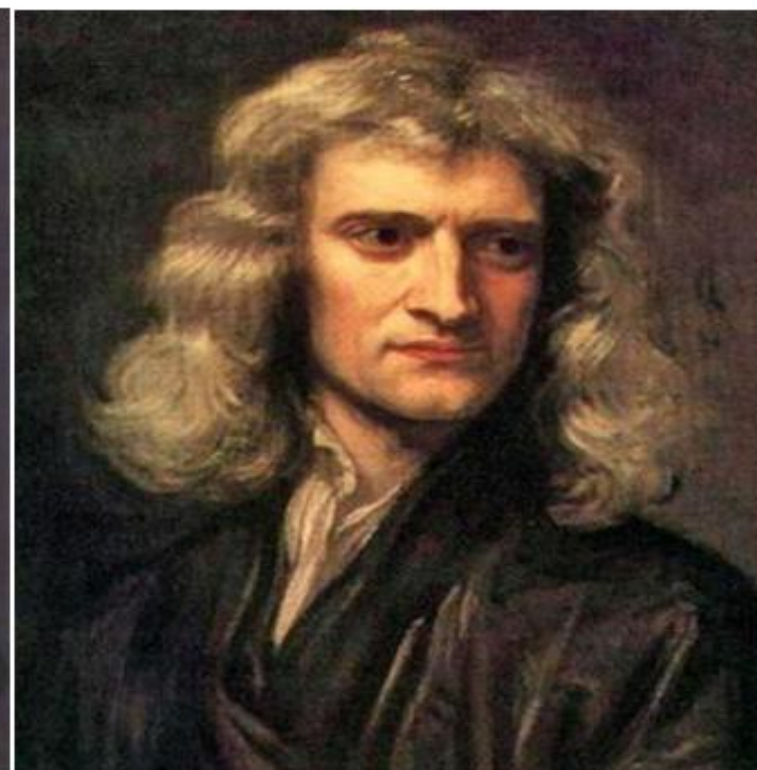
导数符号： $'$ 或 $\frac{d}{dx}$

已知 $y = f(x)$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \frac{d}{dx} y = y' = f'(x) = \frac{d}{dx} f(x) = \frac{df(x)}{dx}$$



戈特弗里德·威廉·莱布尼茨
Gottfried Wilhelm Leibniz
1646-1716



$'$ 或 $\frac{d}{dx}$ 艾萨克·牛顿
Isaac Newton
1643-1727

d) 已知函数 $y = \frac{3}{x}$, $x \neq 0$ 。求 i) $\frac{\Delta y}{\Delta x}$: ii) $\frac{dy}{dx}$: iii) 其在 $x = 2$ 处的导数值。

i) 设 $y = f(x) = \frac{3}{x}$

$$\Delta y = f(x + \Delta x) - f(x)$$

$$= \frac{3}{x + \Delta x} - \frac{3}{x}$$

$$= \frac{3x - 3(x + \Delta x)}{(x + \Delta x)x}$$

$$= \frac{-3\Delta x}{(x + \Delta x)x}$$

$$\frac{\Delta y}{\Delta x} = \frac{\frac{-3\Delta x}{(x + \Delta x)x}}{\Delta x} = \frac{-3}{(x + \Delta x)x}$$

$$ii) \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{-3}{(x + \Delta x)x}$$

$$= \frac{-3}{x^2}$$

导函数, 简称导数

iii) 当 $x = 2$

$$\left. \frac{dy}{dx} \right|_{x=2} = \frac{-3}{4}$$

导数值

习题22.1(2)-2

22.2 微分(differentiate)法则

Pg. 44

$$(C)' = 0, C \text{ 为常数}$$

$$\text{或 } \frac{d}{dx}(C) = 0$$

证明：

$$\text{设 } y = f(x) = C$$

$$(C)' = f'(x)$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{C - C}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{0}{\Delta x}$$

$$= 0 \quad (\text{得证})$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$(x^n)' = nx^{n-1}, n \text{ 为实数}$$

证明:

$$\text{设 } y = f(x) = x^n$$

$$(x^n)' = f'(x)$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\text{或 } \frac{d}{dx}(x^n) = nx^{n-1}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^n - x^n}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(x^n + {}_nC_1 x^{n-1} \Delta x + {}_nC_2 x^{n-2} \Delta x^2 + \dots + {}_nC_n \Delta x^n) - x^n}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x ({}_nC_1 x^{n-1} + {}_nC_2 x^{n-2} \Delta x + \dots + {}_nC_n \Delta x^{n-1})}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} ({}_nC_1 x^{n-1} + {}_nC_2 x^{n-2} \Delta x + \dots + {}_nC_n \Delta x^{n-1})$$

$$= {}_nC_1 x^{n-1}$$

$$= nx^{n-1} \quad (\text{得证})$$

$$(C)' = 0, C \text{ 为常数}$$

$$(x^n)' = nx^{n-1}, n \text{ 为实数}$$

a)求下列各式的导数：

$$i)y = x^{2022}$$

$$ii)y = \sqrt{x}$$

$$iii)y = \pi^2$$

$$iv)y = \frac{1}{\sqrt[3]{x^2}}$$

答案： **【 $2022x^{2021}$ 】** 、 **【 $\frac{1}{2\sqrt{x}}$ 】** 、 **【0】** 、 **【 $\frac{-2}{3\sqrt[3]{x^5}}$ 】**

$$(u \pm v)' = u' \pm v', \quad u, v \text{ 为 } x \text{ 的函数}$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

证明:

$$\text{或 } \frac{d}{dx}(u \pm v) = \frac{d}{dx}u \pm \frac{d}{dx}v$$

已知 $u = u(x)$, $v = v(x)$

$$(u \pm v)' = [u(x) \pm v(x)]'$$

$$= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) \pm v(x + \Delta x) - [u(x) \pm v(x)]}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x) \pm v(x + \Delta x) - v(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x} \pm \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x) - v(x)}{\Delta x}$$

$$= u'(x) \pm v'(x)$$

$$= u' \pm v' \quad (\text{得证})$$

$$(uv)' = u'v + v'u$$

$$(Cu)' = Cu', u \text{ 为 } x \text{ 的函数, } C \text{ 为常数}$$

证明：

已知 $u = u(x)$

$$(Cu)' = [Cu(x)]'$$

$$= \lim_{\Delta x \rightarrow 0} \frac{Cu(x + \Delta x) - Cu(x)}{\Delta x}$$

$$= C \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x}$$

$$= Cu'(x)$$

$$= Cu' \quad (\text{得证})$$

$$\text{或 } \frac{d}{dx}(Cu) = C \frac{d}{dx}u$$

b)求下列各式的导数:

$$i)y = x^8$$

$$ii)y = 2x^{\frac{2}{3}}$$

$$iii)y = x^3 - 3x^2 - 6$$

$$iv)y = x - \frac{2}{x^2}$$

$$v)f(x) = x + \frac{\sqrt{x}}{2} + \sqrt[3]{x}$$

$$vi)f(x) = \sqrt{x} - 3\sqrt{x^3}$$

$$vii)f(x) = (2-x)^2$$

$$viii)f(x) = \frac{x^3 - 2x^2 - 1}{x}$$

$$ix)s = \frac{\sqrt[3]{x} + \sqrt{x} - 1}{\sqrt{x}}$$

答案: $\mathbf{[8x^7]}$ 、 $\mathbf{[\frac{4}{3}x^{-\frac{1}{3}}]}$ 、 $\mathbf{[3x^2 - 6x]}$ 、

$\mathbf{[1 + \frac{4}{x^3}]}$ 、 $\mathbf{[1 + \frac{1}{4\sqrt{x}} + \frac{1}{3\sqrt[3]{x^2}}]}$ 、 $\mathbf{[\frac{1}{2\sqrt{x}} - \frac{9}{2}\sqrt{x}]}$

$\mathbf{[-4 + 2x]}$ 、 $\mathbf{[2x - 2 + \frac{1}{x^2}]}$ 、 $\mathbf{[-\frac{1}{6\sqrt[6]{x^7}} + \frac{1}{2\sqrt{x^3}}]}$

提示: 分母一项 $\rightarrow$ 拆分

$(C)' = 0, C$ 为常数

$(x^n)' = nx^{n-1}$

$(u \pm v)' = u' \pm v'$

c)已知 $f'(2) = 2, g'(2) = -1$, 若 $f(x) = g(x) + 3h(x)$, 求 $h'(2)$ 。

答案: **【1】**

习题22.2a(1)–1(a, b, c, d, e, f, g, h)

挑战题:1.已知 $f(x) = 1 + x + x^2 + x^3 + \dots + x^{100}$, 求 $\lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - f(1)}{\Delta x}$ 。

$(C)' = 0, C$ 为常数

$(x^n)' = nx^{n-1}$

$(u \pm v)' = u' \pm v'$

$$(uv)' = u'v + v'u, \quad u, v \text{ 为 } x \text{ 的函数}$$

口诀：微前保后+微后保前

$$\text{或 } \frac{d}{dx}(uv) = \left(\frac{d}{dx}u \right)v + \left(\frac{d}{dx}v \right)u$$

证明：

已知 $u = u(x)$, $v = v(x)$

$$\begin{aligned} (uv)' &= [u(x)v(x)]' = \lim_{\Delta x \rightarrow 0} \frac{[u(x + \Delta x) \cdot v(x + \Delta x)] - [u(x) \cdot v(x)]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[u(x + \Delta x) \cdot v(x + \Delta x)] + \overbrace{v(x + \Delta x) \cdot u(x) - v(x + \Delta x) \cdot u(x)}^{\text{和为0}} - [u(x) \cdot v(x)]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[u(x + \Delta x) - u(x)]v(x + \Delta x) + [v(x + \Delta x) - v(x)]u(x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x} \lim_{\Delta x \rightarrow 0} v(x + \Delta x) + \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x) - v(x)}{\Delta x} \lim_{\Delta x \rightarrow 0} u(x) \\ &= u'(x)v(x) + v'(x)u(x) \\ &= u'v + v'u \quad (\text{得证}) \end{aligned}$$

$$(uv)' = u'v + v'u$$

d)求下列各式的导数：

$$i)y = (x+1)(x-3)$$

$$ii)y = (2x+3)(x-5)$$

$$iii)y = x^3(4x+5)$$

$$iv)f(x) = (2x+3)(4x^2-6x-5)$$

$$v)f(x) = (x^3+2x^2+1)(4x^2-3)$$

答案： **【 $2x-2$ 】** 、 **【 $4x-7$ 】** 、
【 $x^2(16x+15)$ 】 、 **【 $24x^2-28$ 】** 、
【 $20x^4+32x^3-9x^2-4x$ 】

e) 证明 $(uvw)' = u'vw + uv'w + uvw'$, 其中 u, v, w 为 x 的函数 $(uv)' = u'v + v'u$

据此, 推断 $(abcd)'$ 的结果, 其中 a, b, c, d 为 x 的函数。

已知 $f(x) = (x^2 + 1)(3x - 1)(1 - x^3)$, 求 $f'(2)$ 的值。

习题 22.2 $a(2) - 1(i, j), 2$

挑战题: 1. 已知 $f(x) = (x-1)(x-2)(x-3)\dots(x-2021)$, 求 $f'(2021)$ 的值。

$$\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{v^2}, \quad u, v \text{ 为 } x \text{ 的函数}$$

$$\text{或 } \frac{d}{dx} \left(\frac{u}{v}\right) = \frac{\left(\frac{d}{dx} u\right)v - \left(\frac{d}{dx} v\right)u}{v^2}$$

证明:

已知 $u = u(x)$, $v = v(x)$

口诀: “微上保下-微下保上” 除以 “下” 平方

$$\begin{aligned} \left(\frac{u}{v}\right)' &= \left[\frac{u(x)}{v(x)}\right]' = \lim_{\Delta x \rightarrow 0} \frac{\frac{u(x+\Delta x)}{v(x+\Delta x)} - \frac{u(x)}{v(x)}}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{u(x+\Delta x) \cdot v(x) - u(x) \cdot v(x+\Delta x)}{v(x+\Delta x) \cdot v(x) \cdot \Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{u(x+\Delta x) \cdot v(x) + \overbrace{u(x) \cdot v(x) - u(x) \cdot v(x)}^{\text{和为0}} - u(x) \cdot v(x+\Delta x)}{v(x+\Delta x) \cdot v(x) \cdot \Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[u(x+\Delta x) \cdot v(x) - u(x) \cdot v(x)] - [u(x) \cdot v(x+\Delta x) - u(x) \cdot v(x)]}{v(x+\Delta x) \cdot v(x) \cdot \Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[u(x+\Delta x) - u(x)]v(x) - v(x)[u(x+\Delta x) - u(x)]}{v(x+\Delta x) \cdot v(x) \cdot \Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{v(x)}{v(x+\Delta x) \cdot v(x)} \cdot \underbrace{\lim_{\Delta x \rightarrow 0} \frac{[u(x+\Delta x) - u(x)]}{\Delta x}}_{u'(x)} - \lim_{\Delta x \rightarrow 0} \frac{u(x)}{v(x+\Delta x) \cdot v(x)} \cdot \underbrace{\lim_{\Delta x \rightarrow 0} \frac{[v(x+\Delta x) - v(x)]}{\Delta x}}_{v'(x)} \\ &= \frac{v(x)}{v(x) \cdot v(x)} \cdot u'(x) - \frac{u(x)}{v(x) \cdot v(x)} \cdot v'(x) = \frac{v(x)u'(x) - u(x)v'(x)}{v^2(x)} = \frac{vu' - uv'}{v^2} \quad (\text{得证}) \end{aligned}$$

f)求下列各式的导数:

$$i)y = \frac{3x}{2x+1}$$

$$ii)y = \frac{3x+1}{x-1}$$

$$iii)f(x) = \frac{x^3-2}{x+5}$$

$$iv)f(x) = \frac{3x^2+1}{2x+1}$$

$$v)f(x) = \frac{1}{x-2} + \frac{1}{x+2}$$

$$vi)f(x) = \frac{3x+4}{(5x-2)(2x+1)}$$

答案: $\left[\frac{3}{(2x+1)^2}\right]$ 、 $\left[\frac{-4}{(x-1)^2}\right]$ 、

$$\left[\frac{2x^3+15x^2+2}{(x+5)^2}\right]$$
、 $\left[\frac{6x^2+6x-2}{(2x+1)^2}\right]$ 、

$$\left[\frac{-2x^2-8}{(x-2)^2(x+2)^2}\right]$$
、 $\left[\frac{-30x^2-80x-10}{(5x-2)^2(2x+1)^2}\right]$

$$\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{v^2}$$

习题22.2b-1,3,5,7,9

挑战题:1.已知 $f(x) = \frac{x(x-1)(x-2)\dots(x-2021)}{(x+1)(x+2)\dots(x+2021)}$,求 $f'(0)$ 的值。

复习:

$$(\text{常数})' = 0$$

$$(x^n)' = nx^{n-1}$$

$$(f \pm g)' = f' \pm g'$$

$$(fg)' = f'g + g'f$$
 口诀: 微前保后 + 微后保前

$$(Cf)' = Cf'$$
, C 为常数

$$\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}$$
 口诀: $\frac{\text{微上保下} - \text{微下保上}}{(\text{下})^2}$

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

22.3 链导法

(Chain Rule)

Pg. 52

求 $y = (2x + 1)^2$ 的导数。

解1:

$$\begin{aligned} y &= (2x + 1)^2 \\ &= 4x^2 + 4x + 1 \end{aligned}$$

$$y' = 8x + 4$$

解2:

$$\begin{aligned} y &= (2x + 1)(2x + 1) \\ y' &= (2x + 1)2 + (2x + 1)2 \\ &= 8x + 4 \end{aligned}$$

解3:

$$\begin{aligned} y &= (2x + 1)^2 \\ y' &= 2(2x + 1)2 \\ &= 8x + 4 \end{aligned}$$

求 $y = (2x + 1)^{2021}$ 的导数。

解1:

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$y = (2x + 1)^{2021}$$

$$\text{设 } u = 2x + 1 \quad \text{则 } y = u^{2021}$$

$$\frac{du}{dx} = 2$$

$$\frac{dy}{du} = 2021u^{2020}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= 2021u^{2020} \cdot 2$$

$$= 4042(2x + 1)^{2020}$$

$$\text{解2: } [f(g(x))]' = f'(g(x))g'(x)$$

将 $y = \underbrace{(2x + 1)^{2021}}_{f(x)}$ 看成合成函数，

$$\text{设 } g(x) = 2x + 1, \quad f(x) = x^{2021}$$

$$\therefore f(g(x)) = (2x + 1)^{2021}$$

$$\begin{aligned} g(x) &= 2x + 1 & f(g(x)) &= [g(x)]^{2021} \\ \frac{d}{dx} g(x) &= 2 & \frac{d}{dg(x)} f(g(x)) &= 2021[g(x)]^{2020} \end{aligned}$$

$$\text{则 } [f(g(x))]' = f'(g(x))g'(x)$$

$$= 2021[g(x)]^{2020} \cdot 2$$

$$= 4042(2x + 1)^{2020}$$

$$(u(v(x)))' = u'(v(x)) \cdot v'(x), \quad u, v \text{ 为 } x \text{ 的函数}$$

证明：

$$\text{或 } \frac{d}{dx}(u(v(x))) = \frac{d}{dv}(u(v(x))) \times \frac{d}{dx}(v(x))$$

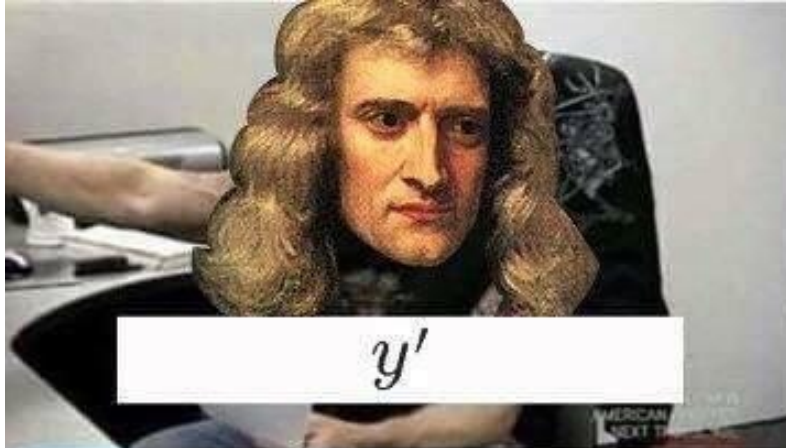
$$\begin{aligned} (u(v(x)))' &= \lim_{\Delta x \rightarrow 0} \frac{[u(v(x + \Delta x))] - [u(v(x))]}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[u(v(x + \Delta x))] - [u(v(x))]}{\Delta x} \cdot \frac{(v(x + \Delta x)) - v(x)}{(v(x + \Delta x)) - v(x)} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[u(v(x + \Delta x))] - [u(v(x))]}{(v(x + \Delta x)) - v(x)} \cdot \frac{(v(x + \Delta x)) - v(x)}{\Delta x} \\ &= \underbrace{\lim_{\Delta x \rightarrow 0} \frac{[u(v(x + \Delta x))] - [u(v(x))]}{(v(x + \Delta x)) - v(x)}}_{u'(v(x))} \cdot \underbrace{\lim_{\Delta x \rightarrow 0} \frac{(v(x + \Delta x)) - v(x)}{\Delta x}}_{v'(x)} \\ &= u'(v(x)) \cdot v'(x) \quad (\text{得证}) \end{aligned}$$

$$[f(g(x))]' = f'(g(x))g'(x)$$

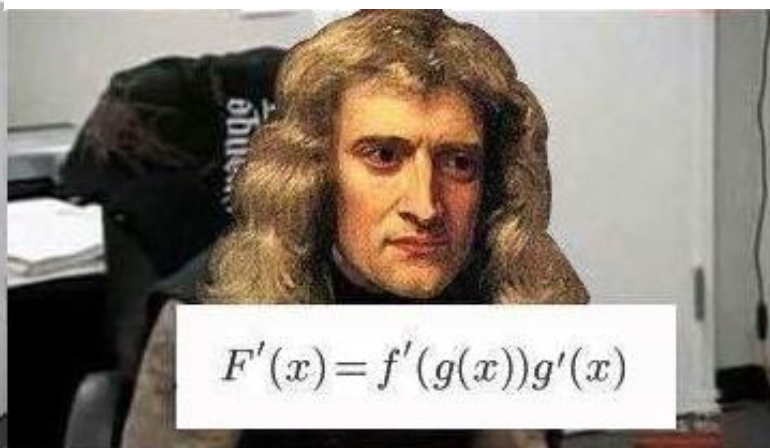
推广 $[f(g(h(x)))]' = ???$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

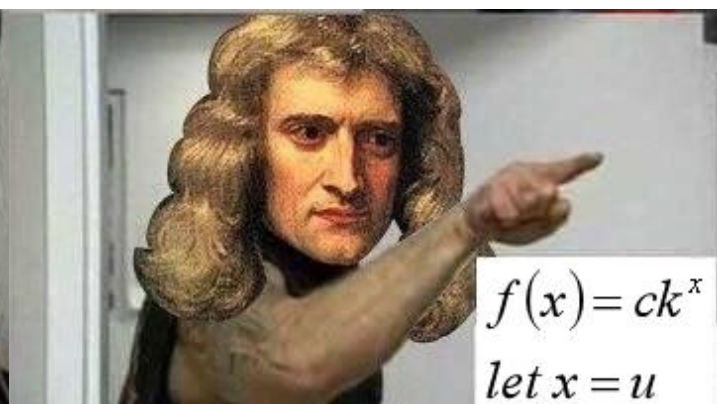
推广 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$



$$y'$$



$$F'(x) = f'(g(x))g'(x)$$



$$f(x) = ck^x$$

let $x = u$



$$\frac{dy}{dx}$$



$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

a) 求下列各式的导数：

$$(x^n)' = nx^{n-1}$$

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

i) $2(3x-1)^7$

$$\begin{aligned} & \left[2(3x-1)^7 \right]', \\ &= (7) 2(3x-1)^6 (3) \\ &= 42(3x-1)^6 \end{aligned}$$

ii) $(x^2 + 1)^4$

$$\begin{aligned} & \left[(x^2 + 1)^4 \right]', \\ &= (4) (x^2 + 1)^3 (2x) \\ &= 8x (x^2 + 1)^3 \end{aligned}$$

iii) $(x^2 + 3x - 1)^2$

$$\begin{aligned} & \left[(x^2 + 3x - 1)^2 \right]', \\ &= (2) (x^2 + 3x - 1)^1 (2x + 3) \\ &= (4x + 6) (x^2 + 3x - 1) \end{aligned}$$

iv) $\sqrt{3x-1}$

$$\begin{aligned} & \left[(3x-1)^{\frac{1}{2}} \right]', \\ &= \left(\frac{1}{2} \right) (3x-1)^{-\frac{1}{2}} (3) \\ &= \frac{3}{2\sqrt{3x-1}} \end{aligned}$$

$$(x^n)' = nx^{n-1}$$

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

a) 求下列各式的导数：

$$v) \frac{1}{4x-3}$$

$$\begin{aligned} & \left[(4x-3)^{-1} \right]', \\ &= -1 (4x-3)^{-2} (4) \\ &= \frac{-4}{(4x-3)^2} \end{aligned}$$

$$vi) \frac{1}{(2x^2 - x - 6)^2}$$

$$\begin{aligned} & \left[(2x^2 - x - 6)^{-2} \right]', \\ &= -2 (2x^2 - x - 6)^{-3} (4x - 1) \\ &= \frac{-8x + 2}{(2x^2 - x - 6)^3} \end{aligned}$$

$$vii) \frac{1}{\sqrt{x^2 + 2x - 4}}$$

$$\begin{aligned} & \left[(x^2 + 2x - 4)^{-\frac{1}{2}} \right]', \\ &= -\frac{1}{2} (x^2 + 2x - 4)^{-\frac{3}{2}} (2x + 2) \\ &= -\frac{x + 1}{\sqrt{(x^2 + 2x - 4)^3}} \end{aligned}$$

b) 已知 $h(x) = f(g(x))$, 且 $h'(2) = 45, f'(8) = 5, g(2) = 8$, 求 $g'(2)$ 的值。

$$h(x) = f(g(x))$$

$$h'(x) = f'(g(x)) \cdot g'(x)$$

当 $x = 2$

$$h'(2) = f'(g(2)) \cdot g'(2)$$

$$45 = f'(8) \cdot g'(2)$$

$$45 = 5 \cdot g'(2)$$

$$g'(2) = 9$$

习题22.3(1)–1, 2, 3, 4

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

c)求下列各式的导数：

$$i)(1-x^2)(x+2)^3$$

$$\begin{aligned} & \left[(1-x^2)(x+2)^3 \right]', \\ &= (-2x)(x+2)^3 + 3(x+2)^2(1) (1-x^2) \end{aligned}$$

同抽 $(x+2)^2$ 抽较小次方项

$$\begin{aligned} &= (x+2)^2 \left[(-2x)(x+2) + 3(1-x^2) \right] \\ &= (x+2)^2 \left[-5x^2 - 4x + 3 \right] \end{aligned}$$

$$ii)(x-2)\sqrt{1-x^2}$$

$$\begin{aligned} & \left[(x-2)(1-x^2)^{\frac{1}{2}} \right]', \\ &= (1)(1-x^2)^{\frac{1}{2}} + \frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x)(x-2) \end{aligned}$$

同抽 $(1-x^2)^{-\frac{1}{2}}$ 抽较小次方项

$$\begin{aligned} &= (1-x^2)^{-\frac{1}{2}} \left[(1-x^2)^1 - x(x-2) \right] \\ &= \frac{-2x^2 + 2x + 1}{\sqrt{1-x^2}} \end{aligned}$$

c)求下列各式的导数:

$$iii) x^2(x-1)(x+2)^2$$

$$\begin{aligned} & [(x^3 - x^2)(x+2)^2]', \\ &= (3x^2 - 2x)(x+2)^2 + 2(x+2)(1)(x^3 - x^2) \end{aligned}$$

同抽 $(x+2)(x)$ 抽较小次方项

$$\begin{aligned} &= x(x+2)[(3x-2)(x+2) + 2(x^2 - x)] \\ &= x(x+2)[5x^2 + 2x - 4] \end{aligned}$$

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

$$iv) \frac{(x+1)^2}{x^3}$$

$$\left[\frac{(x+1)^2}{x^3} \right],$$

$$= \frac{(2)(x+1)x^3 - (3x^2)(x+1)^2}{(x^3)^2}$$

$$= \frac{x^2(x+1)[2x - 3(x+1)]}{x^6}$$

$$= \frac{(x+1)[-x-3]}{x^4}$$

c)求下列各式的导数：

$$v) \frac{3x+2}{\sqrt{x+1}}$$

$$\left[\frac{3x+2}{(x+1)^{\frac{1}{2}}} \right],$$

$$\begin{aligned} &= \frac{(3)(x+1)^{\frac{1}{2}} - \left(\frac{1}{2}\right)(x+1)^{-\frac{1}{2}}(3x+2)}{\left((x+1)^{\frac{1}{2}}\right)^2} \\ &= \frac{(x+1)^{-\frac{1}{2}} \left[(3)(x+1)^1 - \frac{1}{2}(3x+2) \right]}{x+1} \\ &= \frac{3x+4}{2(x+1)\sqrt{x+1}} \end{aligned}$$

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

$$vi) \frac{(2-5x)^2}{x^3-1}$$

$$\left[\frac{(2-5x)^2}{x^3-1} \right],$$

$$\begin{aligned} &= \frac{(2)(2-5x)^1(-5)(x^3-1) - (3x^2)(2-5x)^2}{(x^3-1)^2} \\ &= \frac{(2-5x)[-10(x^3-1) - 3x^2(2-5x)]}{(x^3-1)^2} \\ &= \frac{(2-5x)[5x^3 - 6x^2 + 10]}{(x^3-1)^2} \end{aligned}$$

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

c)求下列各式的导数：

$$vii) \left(\frac{1-x}{1+x} \right)^3$$

$$\begin{aligned} & \left[\left(\frac{1-x}{1+x} \right)^3 \right], \\ &= (3) \left(\frac{1-x}{1+x} \right)^2 \left(\frac{(-1)(1+x) - (1-x)}{(1+x)^2} \right) \\ &= \frac{-6(1-x)^2}{(1+x)^4} \end{aligned}$$

$$viii) x \sqrt{1+x^2}$$

$$\begin{aligned} & \left[x(1+x^2)^{\frac{1}{2}} \right], \\ &= (1)(1+x^2)^{\frac{1}{2}} + \left(\frac{1}{2} \right) (1+x^2)^{-\frac{1}{2}} (2x)(x) \\ &= (1+x^2)^{-\frac{1}{2}} [1+x^2+x^2] \\ &= \frac{2x^2+1}{\sqrt{1+x^2}} \end{aligned}$$

c)求下列各式的导数:

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

$$ix) (x + 3x^2)^4$$

$$\begin{aligned} & \left[(x + 3x^2)^4 \right]' \\ &= (4)(x + 3x^2)^3 (1 + 6x) \\ &= (4 + 24x)(x + 3x^2)^3 \end{aligned}$$

$$x) \sqrt{2x + (x + 3x^2)^4}$$

$$\begin{aligned} & \left[\left(2x + (x + 3x^2)^4 \right)^{\frac{1}{2}} \right]' \\ &= \left(\frac{1}{2} \right) \left(2x + (x + 3x^2)^4 \right)^{-\frac{1}{2}} (2 + (4)(x + 3x^2)^3 (1 + 6x)) \\ &= \frac{1 + (2 + 12x)(x + 3x^2)^3}{\sqrt{2x + (x + 3x^2)^4}} \end{aligned}$$

$$(u(v(x)))' = u'(v(x)) \cdot v'(x)$$

e) 若 $f(x^2 + 1) = \sqrt{4x^4 + 2x^2 + 1}$, 求 $f'(x^2 + 1)$ 。 【 $\frac{4x^2 + 1}{\sqrt{4x^4 + 2x^2 + 1}}$ 】

$$f(x^2 + 1) = (4x^4 + 2x^2 + 1)^{\frac{1}{2}}$$

$$2x \cdot f'(x^2 + 1) = \frac{1}{2} (4x^4 + 2x^2 + 1)^{-\frac{1}{2}} (16x^3 + 4x)$$

$$f'(x^2 + 1) = \frac{4x^2 + 1}{\sqrt{4x^4 + 2x^2 + 1}}$$

f) 若 $f(x^3 - 2) = \sqrt[3]{2x + 5}$, 求 $f'(-3)$ 。 【 $\frac{2}{9 \sqrt[3]{9}}$ 】

综合练习

求下列各函数的导数：

$$a) y = \frac{x^4}{4} - 3x^3 + 5 - 8\pi$$

$$b) y = \frac{2\sqrt{x^5} + 3\sqrt{x^7} - \sqrt{x^9}}{\sqrt{x}}$$

求下列各函数的导数：

$$c) y = (x^2)\left(x^2 + \frac{1}{x} + \frac{1}{x^5}\right)$$

$$d) y = \frac{2}{(2x-3)^2}$$

习题22.3(2) - 5, 7, 9, 10

挑战题: 1. 已知 $f(x)$ 为可微函数, 且 $f\left(\frac{x+1}{x-1}\right) = x$, 求 $f'(0)$ 的值。

挑战题: 2. 已知 $f(x) = 2\left((3x^4 + 5)^6 + 7x^8\right)^9$, 求 $f'(x)$ 的值。

22.4 高阶导数

Pg. 58

若函数 $y=f(x)$ 在 $x=x_0$ 处可导,

则 $f'(x)$ 就称做 $f(x)$ 在 $x=x_0$ 处的一阶导数(简称导数)

即:
$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

同理, 若函数 $y=f(x)$ 的导数 $f'(x)$ 在 $x=x_0$ 处可导

则 $f''(x)$ 就称做 $f(x)$ 在 $x=x_0$ 处的二阶导数

即:
$$f''(x) = \lim_{\Delta x \rightarrow 0} \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

高阶导数的符号：

一阶导数(first derivative)

二阶导数(second derivative)

三阶导数(third derivative)

四阶导数(fourth derivative)

n阶导数(nth derivative)

	$f(x)$	y	
一阶	$f'(x)$	y'	$\frac{dy}{dx}$
二阶	$f''(x)$	y''	$\frac{d^2 y}{dx^2}$
三阶	$f'''(x)$	y'''	$\frac{d^3 y}{dx^3}$
四阶	$f''''(x)$ 或 $f^{(4)}(x)$	y'''' 或 $y^{(4)}$	$\frac{d^4 y}{dx^4}$
...			
n 阶	$f^{(n)}(x)$	$y^{(n)}$	$\frac{d^n y}{dx^n}$

a)求下列各式的二阶导数:

$$i)y = 2x - \frac{x^3}{2}$$

$$\begin{aligned}y &= 2x - \left(\frac{1}{2}\right)x^3 \\y' &= 2 - \left(\frac{1}{2}\right)3x^2 \\y'' &= -\left(\frac{1}{2}\right)3(2)x \\&= -3x\end{aligned}$$

$$ii)y = x + \frac{1}{x}$$

$$\begin{aligned}y &= x + x^{-1} \\y' &= 1 + (-1)x^{-2} \\y'' &= +(-1)(-2)x^{-3} \\&= \frac{2}{x^3}\end{aligned}$$

$$ii)y = \sqrt{1+x^2}$$

$$\begin{aligned}y &= (1+x^2)^{\frac{1}{2}} \\y' &= \frac{1}{2}(1+x^2)^{-\frac{1}{2}}(2x) \\&= x(1+x^2)^{-\frac{1}{2}} \\y'' &= (1+x^2)^{-\frac{1}{2}} + \left(-\frac{1}{2}\right)(1+x^2)^{-\frac{3}{2}}(2x)x \\&= -\frac{x^2}{\sqrt{(1+x^2)^3}} + \frac{1}{\sqrt{1+x^2}}\end{aligned}$$

a)求下列各式的二阶导数:

$$iv) y = \frac{1}{\sqrt{x-1}}$$

$$y = (x-1)^{-\frac{1}{2}}$$

$$y' = -\frac{1}{2}(x-1)^{-\frac{3}{2}}(1)$$

$$y'' = -\frac{1}{2}\left(-\frac{3}{2}\right)(x-1)^{-\frac{5}{2}}(1) \\ = \frac{3}{4\sqrt{(x-1)^5}}$$

$$v) y = \frac{x}{\sqrt{1-x^2}}$$

$$y = x(1-x^2)^{-\frac{1}{2}}$$

$$y' = (1-x^2)^{-\frac{1}{2}} + \left(-\frac{1}{2}\right)(1-x^2)^{-\frac{3}{2}}(-2x)x$$

$$= (1-x^2)^{-\frac{1}{2}} + x^2(1-x^2)^{-\frac{3}{2}}$$

$$= (1-x^2)^{-\frac{3}{2}} \left[(1-x^2)^1 + x^2 \right]$$

$$= (1-x^2)^{-\frac{3}{2}}$$

$$y'' = \left(-\frac{3}{2}\right)(1-x^2)^{-\frac{5}{2}}(-2x) = \frac{3x}{\sqrt{(1-x^2)^5}}$$

b)求 $y = \frac{2}{x}$ 的一阶、二阶、三阶导数，据此，求 n 阶导数。

$$y = 2x^{-1}$$

$$y' = -2x^{-2}$$

$$y'' = -2(-2)x^{-3}$$

$$y''' = -2(-2)(-3)x^{-4}$$

$$y^{(4)} = -2(-2)(-3)(-4)x^{-5}$$

...

$$y^{(n)} = -2(-2)(-3)(-4)\dots(-n)x^{-(n+1)}$$

$$\therefore y' = -\frac{2}{x^2}, y'' = \frac{4}{x^3}, y''' = -\frac{12}{x^4}, y^{(n)} = (-1)^n \frac{2(n!)}{x^{n+1}}$$

c)已知 $y = 2x^3 - 3x^2 - 6x + 10$,求 y', y'', y''' , 在 $x = 1$ 处的值。

$$y = 2x^3 - 3x^2 - 6x + 10$$

$$y' = 6x^2 - 6x - 6 \Rightarrow y'|_{x=1} = 6(1)^2 - 6(1) - 6 = -6$$

$$y'' = 12x - 6 \Rightarrow y''|_{x=1} = 12(1) - 6 = 6$$

$$y''' = 12 \Rightarrow y'''|_{x=1} = 12$$

d)已知 $y = \frac{10-15x^2}{x}$, 求 $\frac{dy}{dx}, \frac{d^2y}{dx^2}$, 据此, 证明 $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 30 = 0$ 。

$$y = \frac{10-15x^2}{x}$$

$$= 10x^{-1} - 15x$$

$$\frac{dy}{dx} = -10x^{-2} - 15$$

$$\frac{d^2y}{dx^2} = 20x^{-3}$$

$$\text{左式} = x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 30$$

$$= x(20x^{-3}) + 2(-10x^{-2} - 15) + 30$$

$$= \frac{20}{x^2} - \frac{20}{x^2} - 30 + 30$$

$$= 0$$

$$= \text{右式(得证)}$$

习题22.4-3, 7, 9, 10

挑战题: 1. 已知 $f(x)$ 为三次多项式, 且 $f(1) = 13, f'(1) = 25, f''(1) = 34, f'''(1) = 24$, 求 $f(x)$ 。

挑战题: 2. 已知 $f(x) = -\frac{x}{x+1}$, 求 $\frac{d^n f(x)}{dx^n}$ 。